

CAMPUS MANAGEMENT SYSTEM

*Project report submitted in fulfilment of the
requirements for the Degree of*

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

By

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Under The Guidance Of

Dr. Abhishek Bhattacharjee

Subject

INT327: Cloud Infrastructure and Resource Management



School Of Computer Science and Engineering

Lovely Professional University

Phagwara, Punjab (India)

Project Introduction

Overview

This project demonstrates the deployment of a complete enterprise-grade cloud infrastructure on Microsoft Azure for a Smart University Campus. The system is designed to manage Admissions, Financial Aid, Registrar, Faculty, IT Administration, Library, Student Affairs, and Alumni departments — each running on isolated virtual networks with dedicated virtual machines, role-based access control, and security policies.

Technologies Used

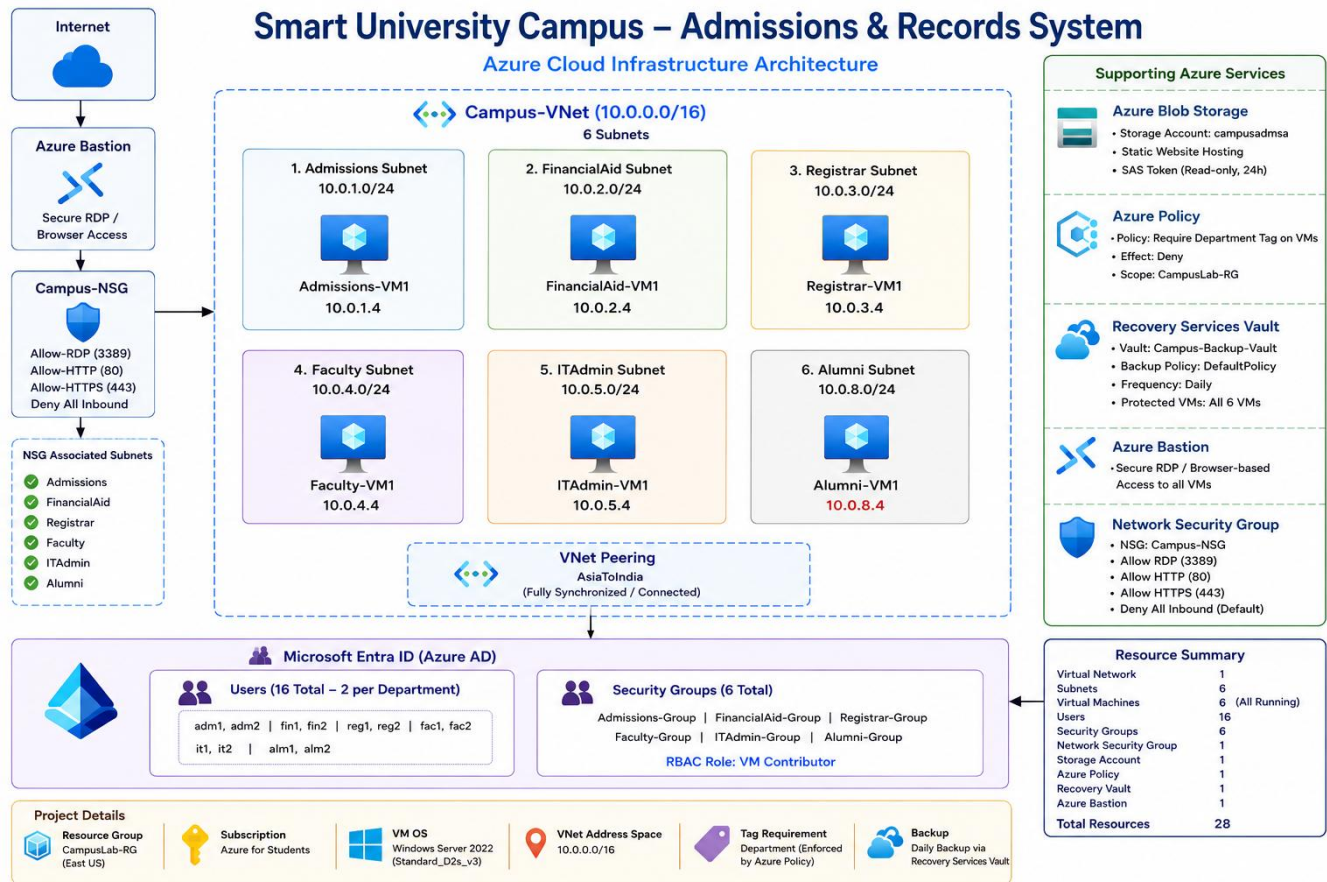
Technology	Usage
Microsoft Azure	Cloud platform for all infrastructure deployment
Azure Virtual Network	Campus-VNet with 8 subnets (10.0.0.0/16)
Azure Virtual Machines	Windows Server 2022, Standard_D2s_v3
Microsoft Entra ID	16 users, 8 groups, RBAC assignments
Azure Blob Storage	Static website hosting, SAS token generation
Azure Policy	Custom deny policy for Department tag enforcement
Recovery Services Vault	Daily VM backup with DefaultPolicy
Azure Bastion	Secure RDP/browser-based VM access

Architecture

Infrastructure Summary

Component	Resource Name	Details
Resource Group	CampusLab-RG	East US · Azure for Students
Virtual Network	Campus-VNet	10.0.0.0/16 · 8 Subnets
Subnets	Admissions, FinancialAid, Registrar, Faculty, ITAdmin, Library, StudentAffairs, Alumni	10.0.1.0/24 to 10.0.8.0/24
NSG	Campus-NSG	RDP, HTTP, HTTPS + Deny rules
Virtual Machines	Admissions-VM1, Faculty-VM1, FinancialAid-VM1, Registrar-VM1	Windows Server 2022
Users	16 Users (2 per dept)	Entra ID
Groups	8 Security Groups	VM Contributor RBAC
Storage Account	campusadmsa	LRS · Static Website · SAS Token
Custom Policy	Require Department Tag on VMs	Deny effect · CampusLab-RG
Backup Vault	Campus-Backup-Vault	DefaultPolicy · Daily · Admissions-VM1

Architecture Diagram



Screen Recording

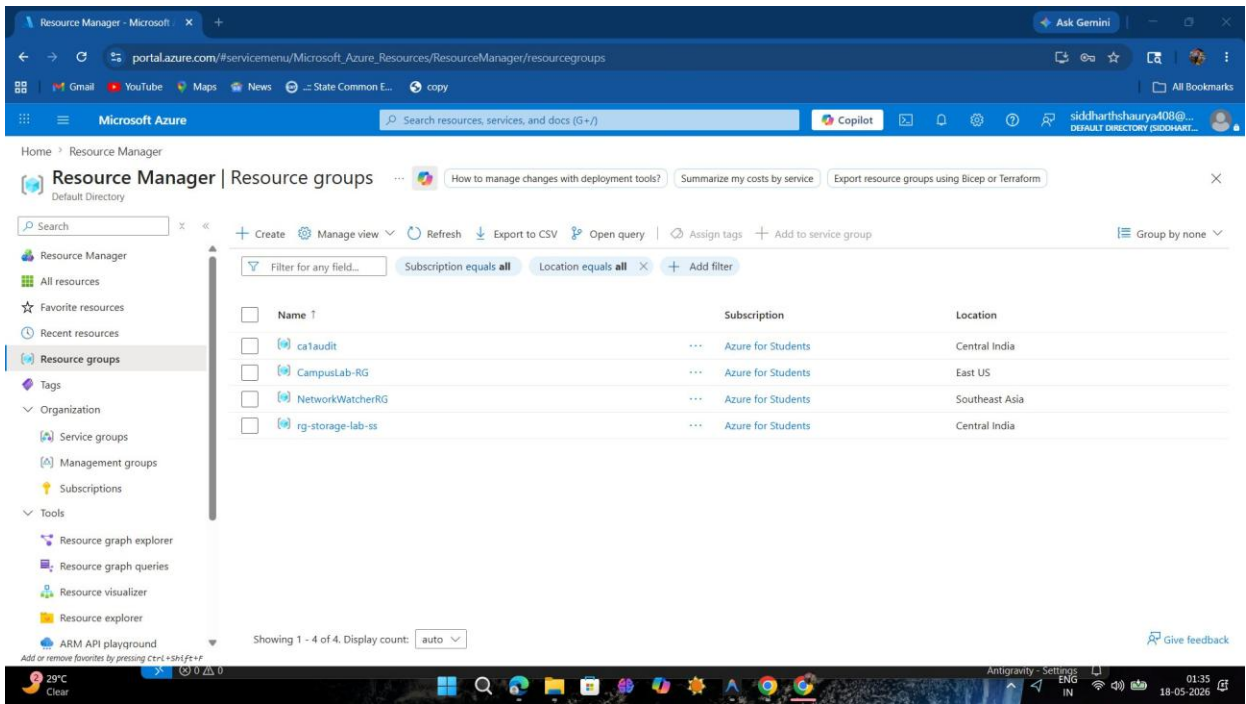
The following Google Drive folder contains screen recordings demonstrating the complete Azure infrastructure deployment, VM connectivity tests, webpage access from VM, and all project components.

Google Drive Link:

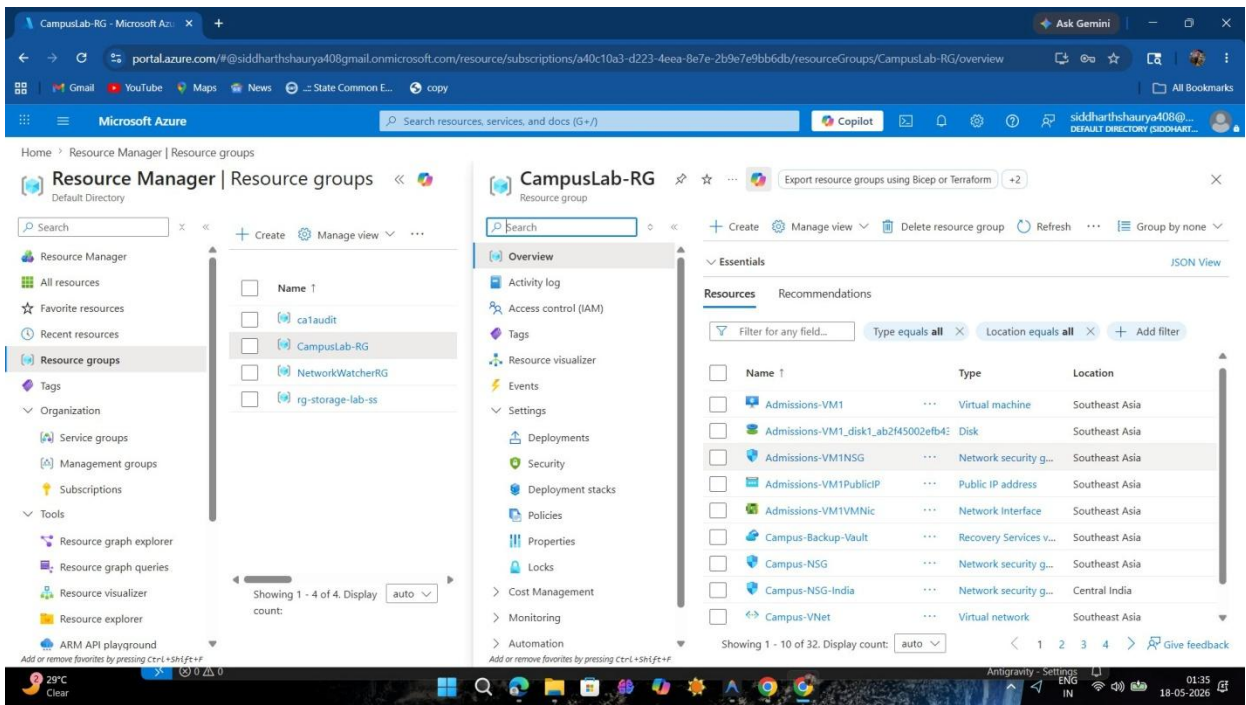
<https://drive.google.com/drive/folders/1MTuv4yqWK-jjr6b4AfQEYrAtuHugfrfT?usp=sharing>

Screenshots: Main Infrastructure

Screenshot 1 — Resource Groups Overview



Screenshot 2 — Campus Lab-RG — All Resources (32 Resources)



Screenshot 3 — Campus-VNet — Virtual Network Overview

The screenshot displays the Microsoft Azure portal interface for the 'Campus-VNet' virtual network. The page is titled 'Campus-VNet' and includes a search bar and navigation links. The left sidebar shows the 'Overview' tab selected, with options like 'Activity log', 'Access control (IAM)', 'Tags', 'Diagnose and solve problems', 'Resource visualizer', 'Settings', 'Monitoring', 'Automation', and 'Help'. The main content area shows the 'Essentials' section with details about the resource group, location, subscription, and address space. Below this, the 'Capabilities' section is visible, showing 'DDoS protection', 'Azure Firewall', 'Peerings', and 'Microsoft Defender for Cloud'. The 'Peerings' section indicates that there is 1 peering configured.

Microsoft Azure

Home > Resource Manager | Resource groups > CampusLab-RG

Campus-VNet Virtual network

Search resources, services, and docs (G+J)

Ask Gemini

sidharthchaurya408@...
DEFAULT DIRECTORY (SIDHART...)

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

Automation

Help

Essentials

Resource group (move) : CampusLab-RG

Location (move) : Southeast Asia

Subscription (move) : Azure for Students

Subscription ID : a40c10a3-d223-4eea-8e7e-2b9e7e9bb6db

Address space : 10.0.0.0/16

Subnets : 8 subnets

DNS servers : Azure provided DNS service

BGP community string : Configure

Virtual network ID : 8be402dd-9208-458c-8fb8-a1a0525d917d

Tags (edit) : Add tags

Topology Properties Capabilities Recommendations Tutorials

DDoS protection

Configure additional protection from distributed denial of service attacks.

Not configured

Azure Firewall

Protect your network with a stateful L3-L7 firewall.

Not configured

Peerings

Seamlessly connect two or more virtual networks.

1 peering

Microsoft Defender for Cloud

Strengthen the security posture of your environment.

29°C Clear

Antigravity - Settings ENG IN

01:36 18-05-2026

Screenshot 4 — Campus-VNet — VNet Peering

The screenshot displays the Microsoft Azure portal interface for the 'Campus-VNet' virtual network, specifically the 'Peerings' section. The page is titled 'Campus-VNet | Peerings' and includes a search bar and navigation links. The left sidebar shows the 'Peerings' tab selected, with options like 'Address space', 'Connected devices', 'Subnets', 'Bastion', 'DDoS protection', 'Firewall', 'Microsoft Defender for Cloud', 'Network manager', 'DNS', 'Service endpoints', 'Private endpoints', and 'Properties'. The main content area shows a table of virtual network peerings. The table has columns for 'Name', 'Peering sync status', 'Peering state', 'Remo...', 'Virtu...', and 'Cross-tenant'. One peering is listed: 'AsiaToIndia' with a 'Fully Synchronized' status and a 'Connected' state.

Microsoft Azure

Home > Resource Manager | Resource groups > CampusLab-RG > Campus-VNet

Campus-VNet | Peerings Virtual network

Search resources, services, and docs (G+J)

Ask Gemini

sidharthchaurya408@...
DEFAULT DIRECTORY (SIDHART...)

Peerings

Virtual network peering enables you to seamlessly connect two or more virtual networks in Azure. The virtual networks appear as one for connectivity purposes. [Learn more](#)

Filter by name...

Showing all 1 items

Name	Peering sync status	Peering state	Remo...	Virtu...	Cross-tenant
AsiaToIndia	Fully Synchronized	Connected	Campus...	Disabled	No

29°C Clear

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01:40 18-05-2026

Screenshot 5 — Campus-NSG — Inbound Security Rules

Home > Resource Manager | Resource groups > CampusLab-RG > Campus-NSG

Campus-NSG | Inbound security rules

Network security group

Search resources, services, and docs (G+)

Filter by name

Port == all Protocol == all Source == all Destination == all Action == all

Priority	Name	Port	Protocol	Source	Destination	Action
100	Allow-RDP	3389	Tcp	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Inbound security rules
Outbound security rules
Network interfaces
Subnets
Properties
Locks
Monitoring
Automation
Help

29°C Clear 01:42 18-05-2026

Screenshot 6 — Campus-NSG — All 8 Subnets Associated

Home > Resource Manager | Resource groups > CampusLab-RG > Campus-NSG

Campus-NSG | Subnets

Network security group

Search resources, services, and docs (G+)

Associate

Search subnets

Name	Address range	Virtual network
Admissions	10.0.1.0/24	Campus-VNet
Alumni	10.0.8.0/24	Campus-VNet
Faculty	10.0.4.0/24	Campus-VNet
FinancialAid	10.0.2.0/24	Campus-VNet
ITAdmin	10.0.5.0/24	Campus-VNet
Library	10.0.6.0/24	Campus-VNet
Registrar	10.0.3.0/24	Campus-VNet
StudentAffairs	10.0.7.0/24	Campus-VNet

Overview
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Screenshot 7 — Virtual Machines — All 4 VMs Running

The screenshot shows the Microsoft Azure portal interface. The left sidebar displays the navigation menu with 'Virtual machines' selected. The main content area shows a list of virtual machines under the 'Virtual machines' tab. The table lists 4 VMs, all with a status of 'Running'.

Name	Subscription	Resource Group	Location	Status	Operating system	Size	Public IP address	Disks
Admissions-VM1	Azure for Stude...	CampusLab-RG	Southeast Asia	Running	Windows	Standard_D2s_v3	4.193.208.103	1
Faculty-VM1	Azure for Stude...	CampusLab-RG	Central India	Running	Windows	Standard_D2s_v3	52.172.226.150	1
FinancialAid-VM1	Azure for Stude...	CampusLab-RG	Southeast Asia	Running	Windows	Standard_D2s_v3	4.193.120.16	1
Registrar-VM1	Azure for Stude...	CampusLab-RG	Southeast Asia	Running	Windows	Standard_D2s_v4	52.237.127.39	1

Screenshot 8 — Admissions-VM1 — Department Tag Applied

The screenshot shows the Microsoft Azure portal interface for the 'Admissions-VM1' virtual machine. The left sidebar displays the navigation menu with 'Virtual machines' selected. The main content area shows the 'Tags' tab for the 'Admissions-VM1' virtual machine. The 'Tags' tab displays a list of tags, including 'Department: Admissions'.

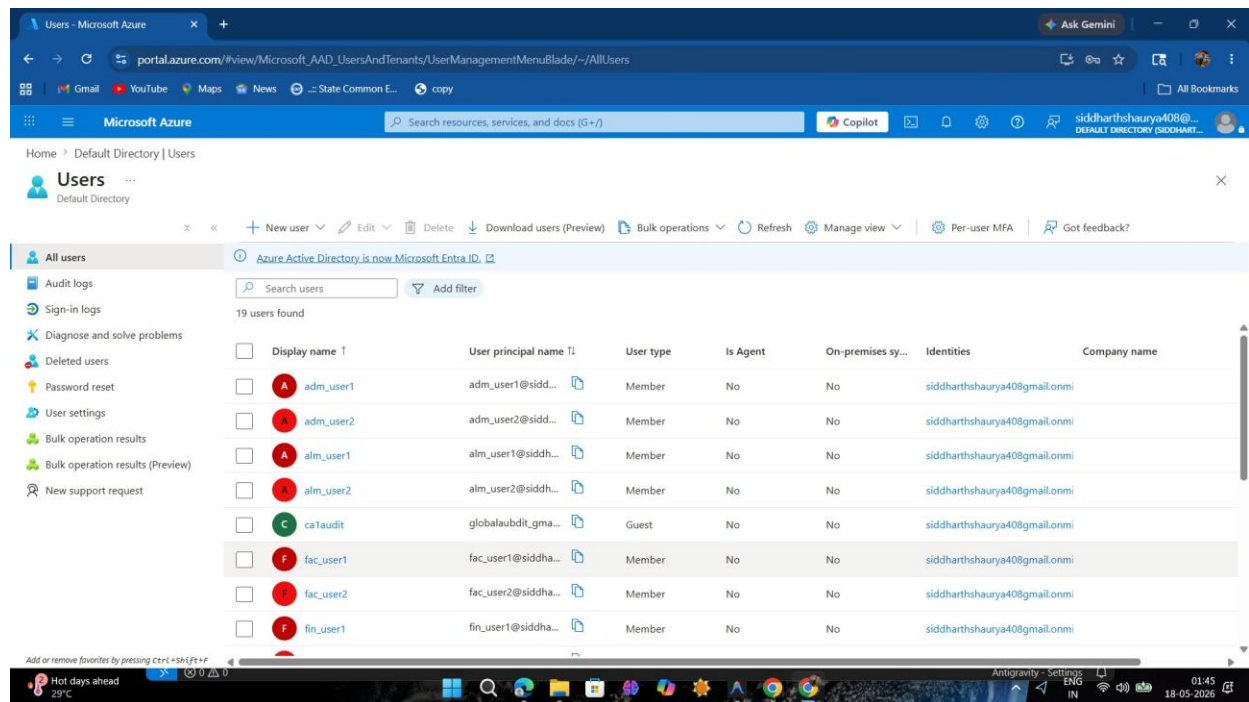
Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. Tag names are case insensitive, but tag values are case sensitive. [Learn more about tags.](#)

Do not enter names or values that could make your resources less secure or that contain personal/sensitive information because tag data will be replicated globally.

Name	Value
Department	Admissions

Admissions-VM1 (Virtual machine)
Department: Admissions
No changes

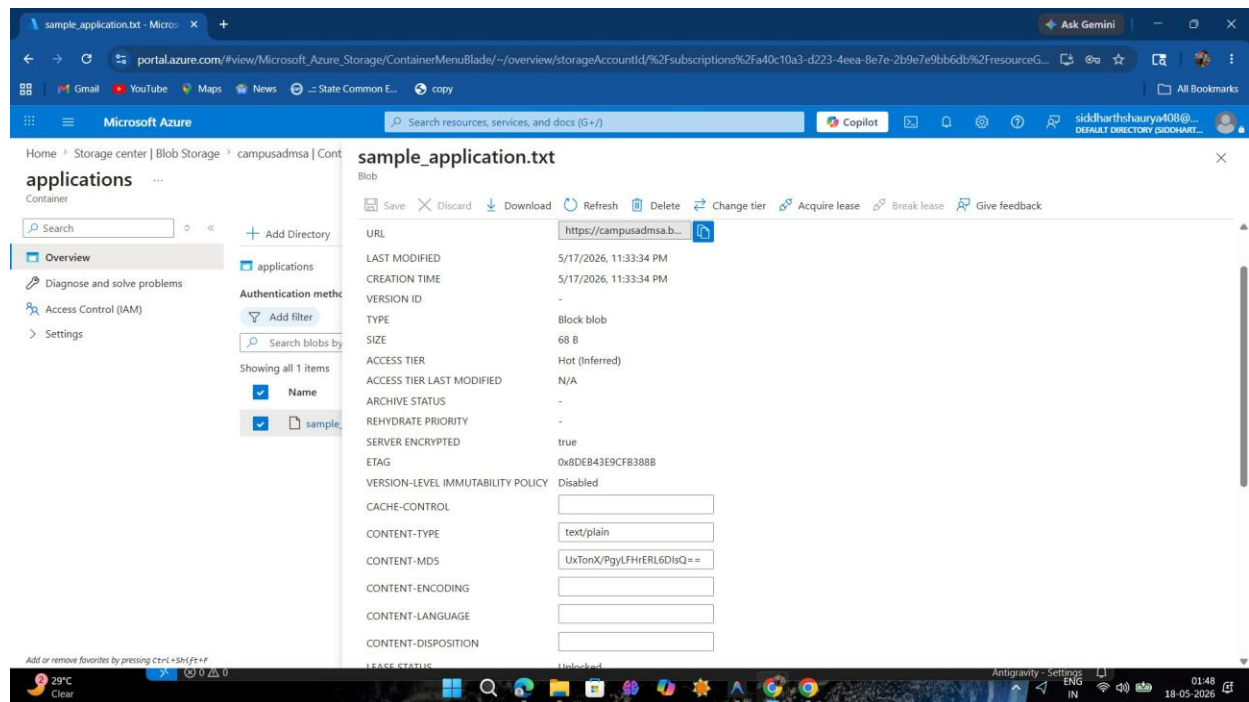
Screenshot 9 — Entra ID — 16 Users Created



The screenshot shows the Microsoft Azure portal interface for the 'Default Directory' under 'Users'. A notification banner at the top states 'Azure Active Directory is now Microsoft Entra ID'. The left sidebar contains navigation options: All users, Audit logs, Sign-in logs, Diagnose and solve problems, Deleted users, Password reset, User settings, Bulk operation results, Bulk operation results (Preview), and New support request. The main area displays a table of 19 users found. The table columns are: Display name, User principal name, User type, Is Agent, On-premises sy..., Identities, and Company name. The users listed are:

Display name	User principal name	User type	Is Agent	On-premises sy...	Identities	Company name
adm_user1	adm_user1@siddh...	Member	No	No	siddharthshaurya408@gmail.onmi	
adm_user2	adm_user2@siddh...	Member	No	No	siddharthshaurya408@gmail.onmi	
alm_user1	alm_user1@siddh...	Member	No	No	siddharthshaurya408@gmail.onmi	
alm_user2	alm_user2@siddh...	Member	No	No	siddharthshaurya408@gmail.onmi	
ca1audit	globalaudit_gma...	Guest	No	No	siddharthshaurya408@gmail.onmi	
fac_user1	fac_user1@siddh...	Member	No	No	siddharthshaurya408@gmail.onmi	
fac_user2	fac_user2@siddh...	Member	No	No	siddharthshaurya408@gmail.onmi	
fin_user1	fin_user1@siddh...	Member	No	No	siddharthshaurya408@gmail.onmi	

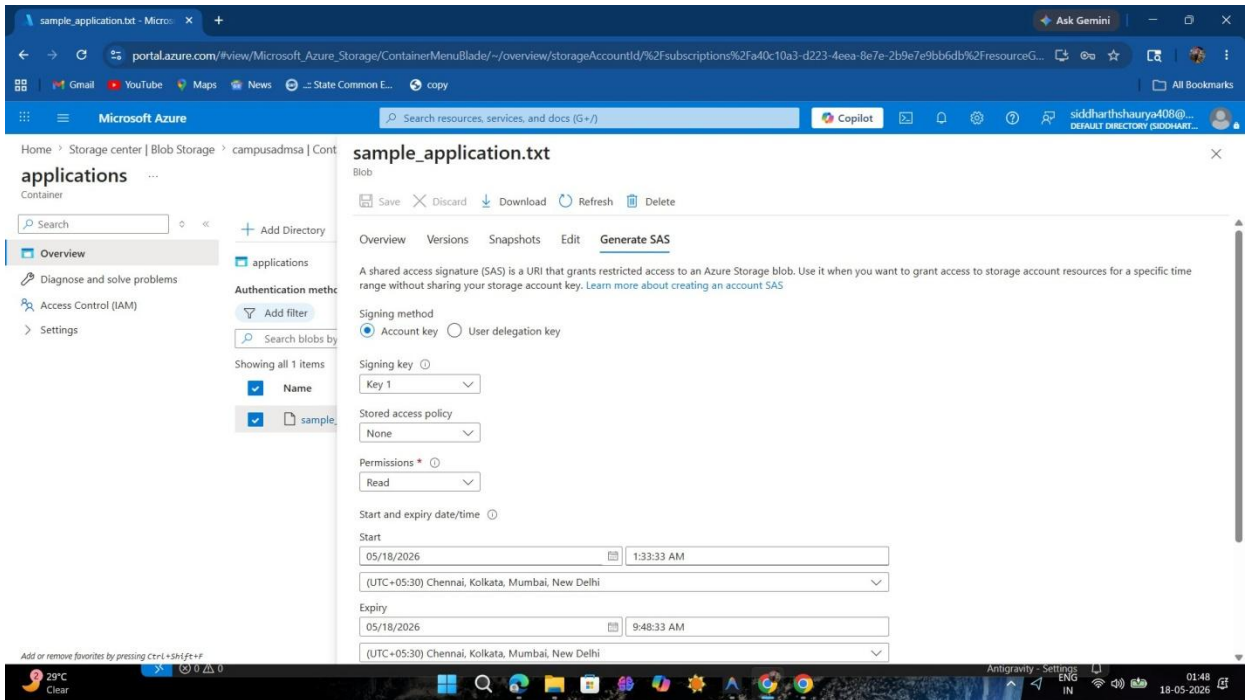
Screenshot 10 — Storage Account — Blob Container with File



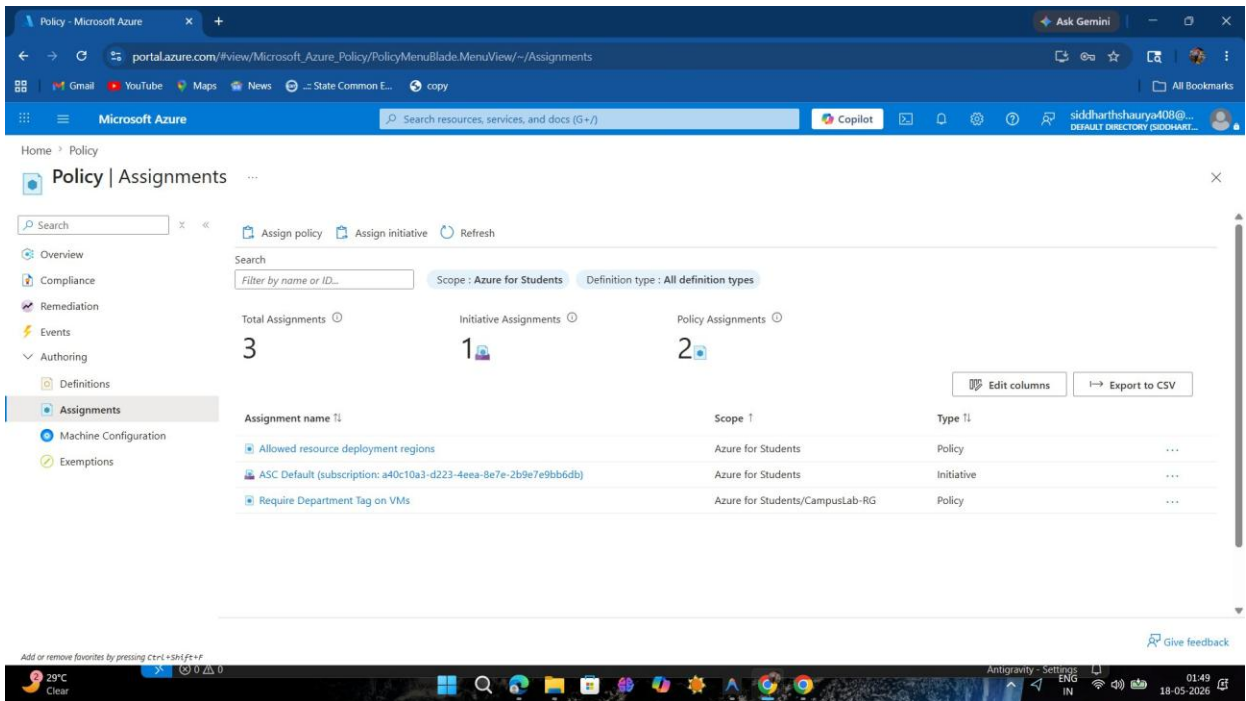
The screenshot shows the Microsoft Azure portal interface for a Storage Account named 'campusadmsa'. The left sidebar contains navigation options: applications, Diagnose and solve problems, Access Control (IAM), and Settings. The main area displays the 'sample_application.txt' Blob container. The container details are shown in a table:

Property	Value
URL	https://campusadmsa.b...
LAST MODIFIED	5/17/2026, 11:33:34 PM
CREATION TIME	5/17/2026, 11:33:34 PM
VERSION ID	-
TYPE	Block blob
SIZE	68 B
ACCESS TIER	Hot (inferred)
ACCESS TIER LAST MODIFIED	N/A
ARCHIVE STATUS	-
REHYDRATE PRIORITY	-
SERVER ENCRYPTED	true
ETAG	0x8DEB43E9CFB388B
VERSION-LEVEL IMMUTABILITY POLICY	Disabled
CACHE-CONTROL	
CONTENT-TYPE	text/plain
CONTENT-MD5	UxTonX/PgyLFHrERL6DlsQ==
CONTENT-ENCODING	
CONTENT-LANGUAGE	
CONTENT-DISPOSITION	

Screenshot 11 — SAS Token Generation



Screenshot 12 — Azure Policy — Assignments Dashboard



Screenshot 13 — Recovery Services Vaults List

The screenshot shows the Microsoft Azure portal interface. The browser address bar displays `portal.azure.com/#browse/Microsoft.RecoveryServices%2Fvaults`. The page title is "Recovery Services vaults". Below the title, there are search filters: "List vaults with failed backup jobs in the last 24 hours", "Find vaults with data sources missing RP in the last 7 days", and a question "What are the benefits and applications of Azure Recovery Services?". The page includes a "Create" button and a "Manage Deleted Vaults" link. A filter bar shows "Subscription equals all", "Resource Group equals all", "Location equals all", and "Security level equals all". The table below lists the vaults:

Name	Type	Security level	Resource Group	Location	Subscription
Campus-Backup-Vault	Recovery Services vault	Poor	CampusLab-RG	Southeast Asia	Azure for Students
vault-moiusz7n	Recovery Services vault	Poor	rg-storage-lab-ss	Central India	Azure for Students

Showing 1 - 2 of 2. Display count: auto

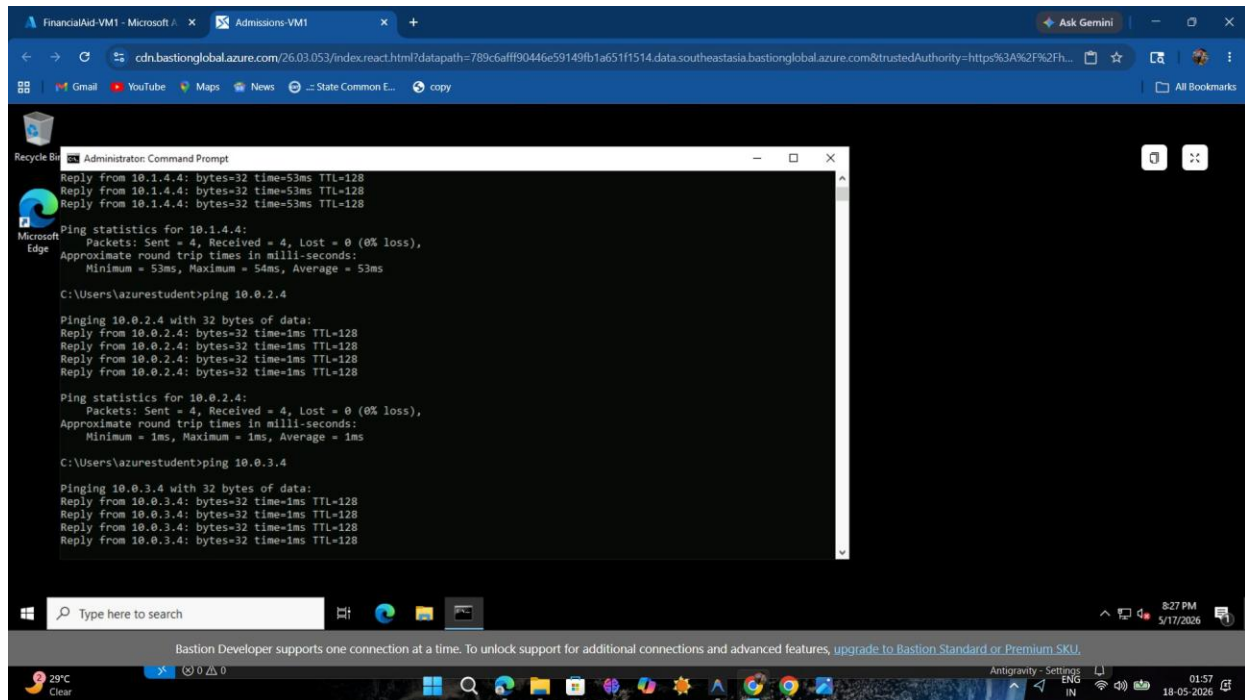
Screenshot 14 — Backup Items — Admissions-VM1 Protected

The screenshot shows the Microsoft Azure portal interface for the "Backup Items (Azure Virtual Machine)" page. The browser address bar displays `portal.azure.com/#view/Microsoft_Azure_DataProtection/V1ProtectedItemsListBlade/vaultId/%2Fsubscriptions%2Ffa40c10a3-d223-4eea-8e7e-2b9e7e9bb6db%2FresourceGroups%2F...`. The page title is "Backup Items (Azure Virtual Machine)". Below the title, there are links for "Refresh", "Add", "Filter", "Change policy", "Bulk restore (Preview)", and "Feedback". A search bar contains the text "Show me the virtual machines that can be protected in this vault". Below the search bar, there are two informational messages: "You can now back up VMs across subscriptions with ease. Centralize protection and simplify multisubscription governance. Learn more." and "Accelerate and simplify large-scale recovery by restoring multiple VMs with Bulk Restore (Public Preview). Learn more." The table below lists the backup items:

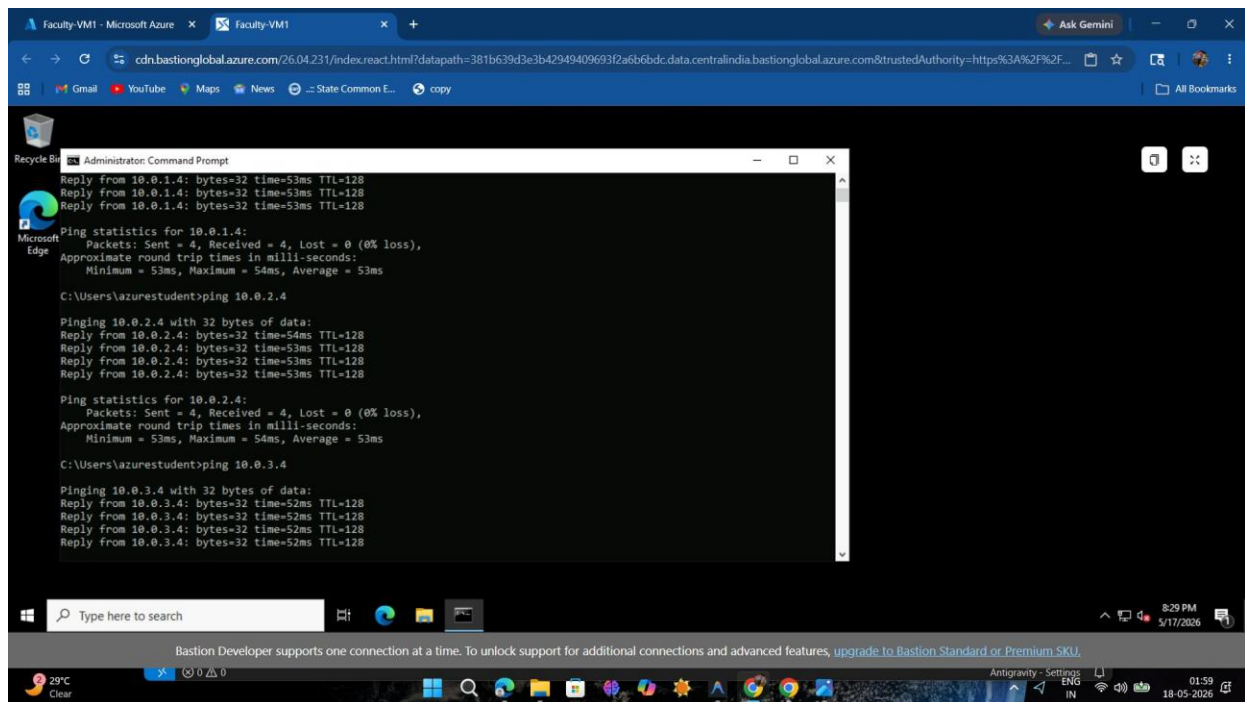
Name	Subscription	Resource Group	Backup Pre-Check	Last Backup Status	Source scan integra...	Source scan summary	Latest restore poi...	Policy name	Policy sub typ
Admissions-VM1	Azure for Students	CampusLab-RG	Passed	Warning (Initial backup pending)	Not configured	-	-	DefaultPolicy	Standard

< Previous Page 1 of 1 Next >

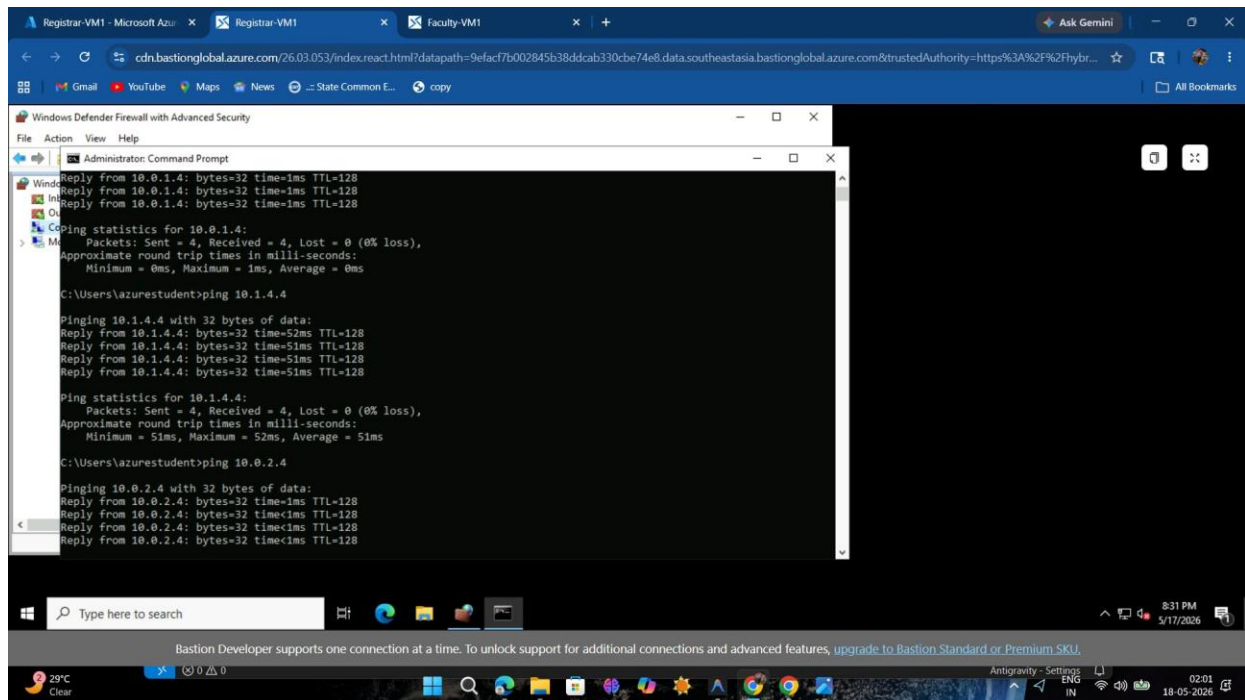
Screenshot 15 — VM Ping Test — Admissions-VM1 (Successful)



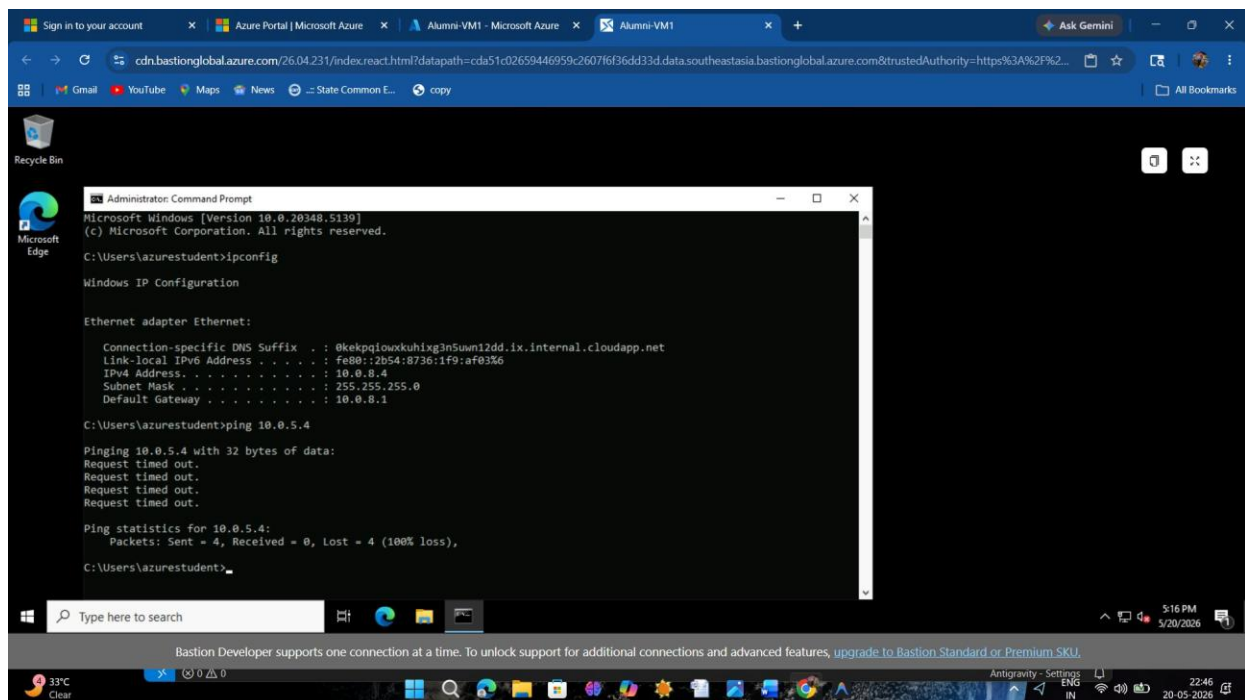
Screenshot 16 — VM Ping Test — Faculty-VM1 (Successful)



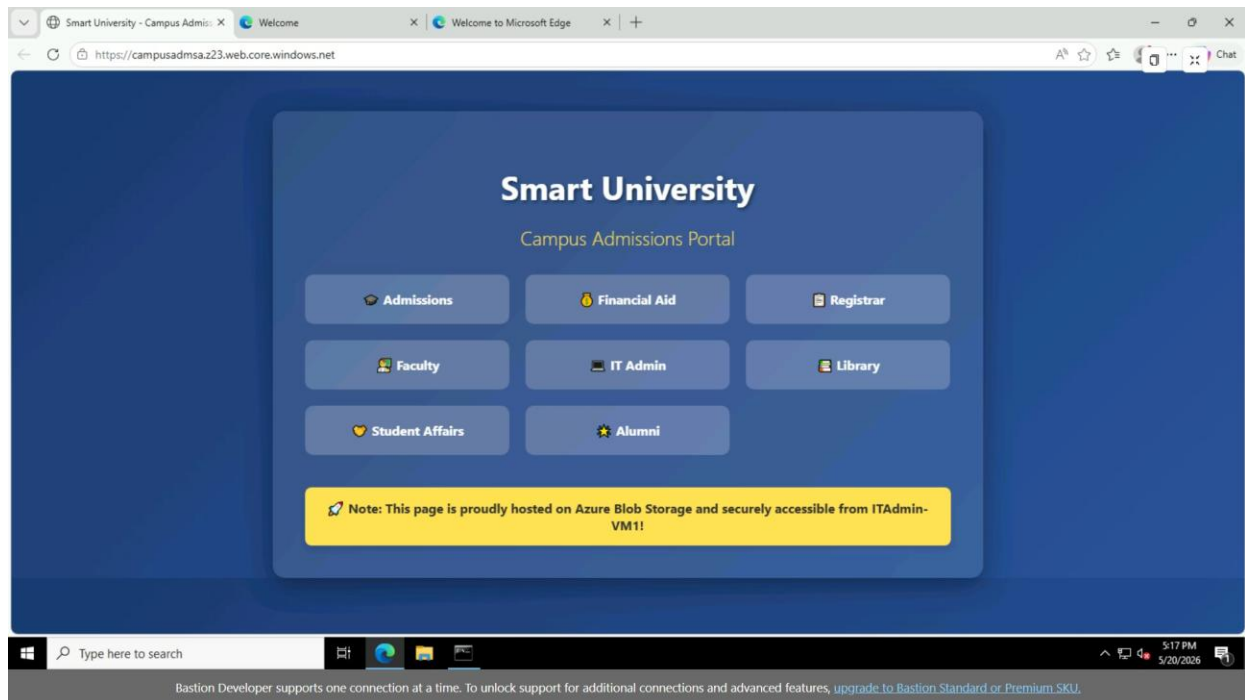
Screenshot 17 — VM Ping Test — Registrar-VM1 (Successful)



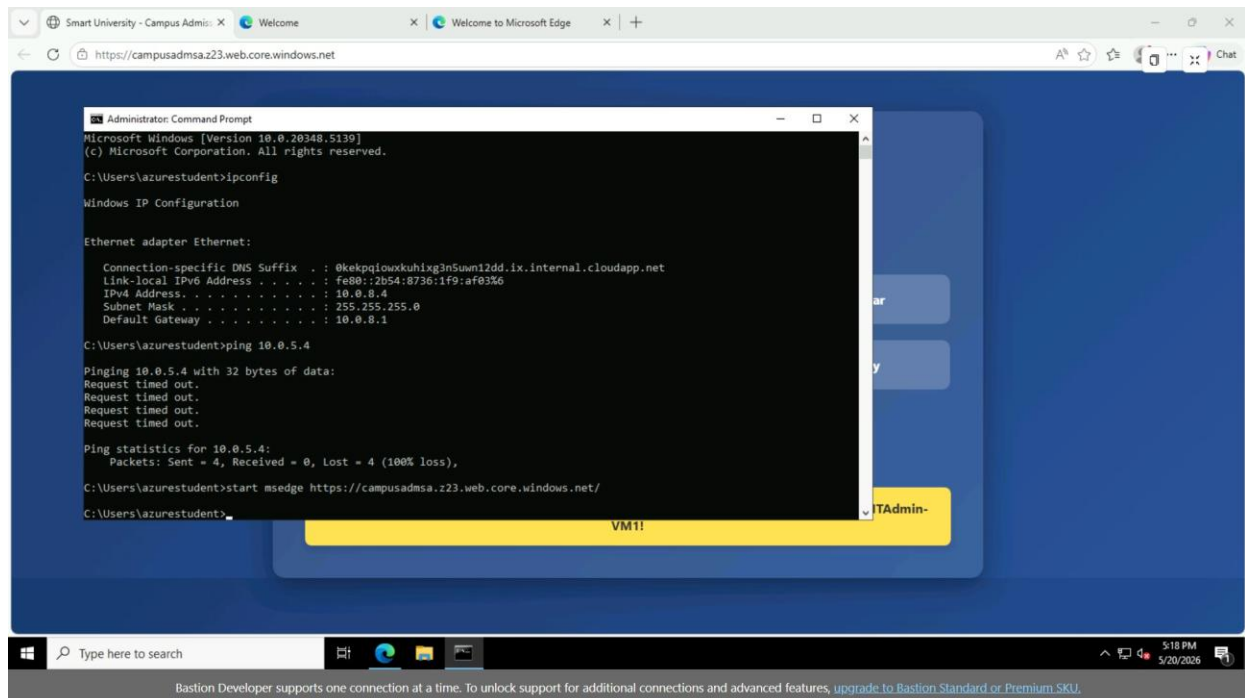
Screenshot 18 — CampusLab-RG — New Deployment Resources



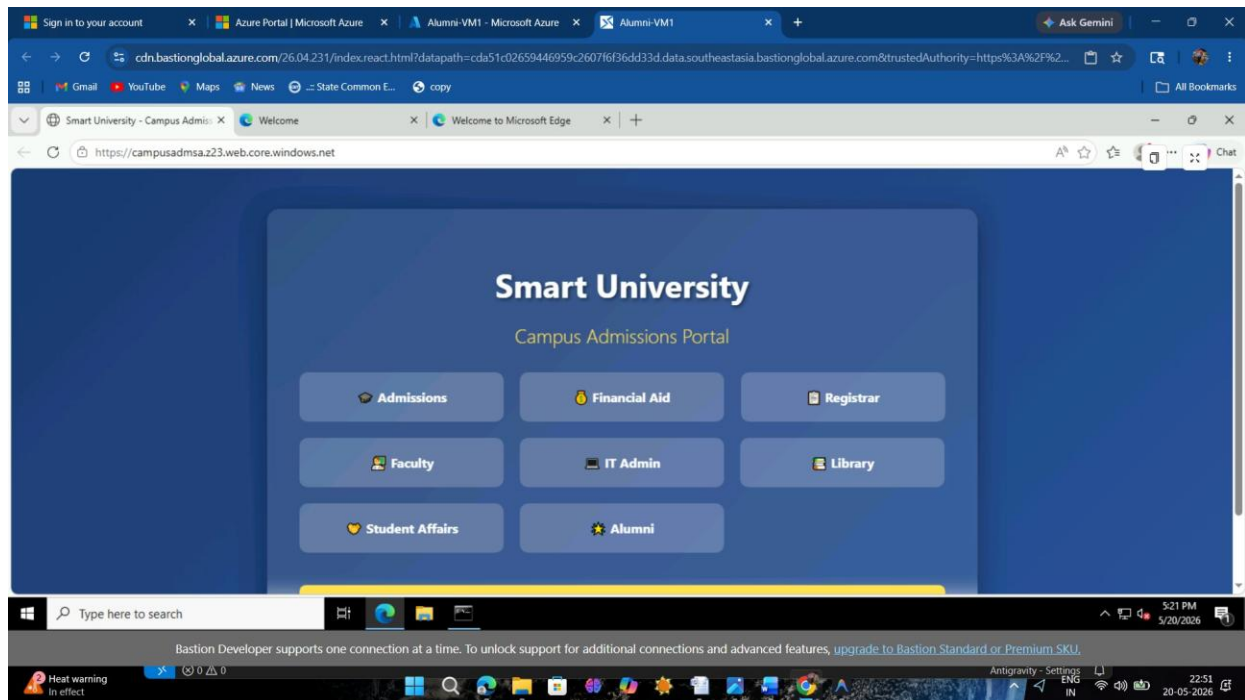
Screenshot 19 — Alumni-VM1 — Ping to ITAdmin Blocked (100% Loss)



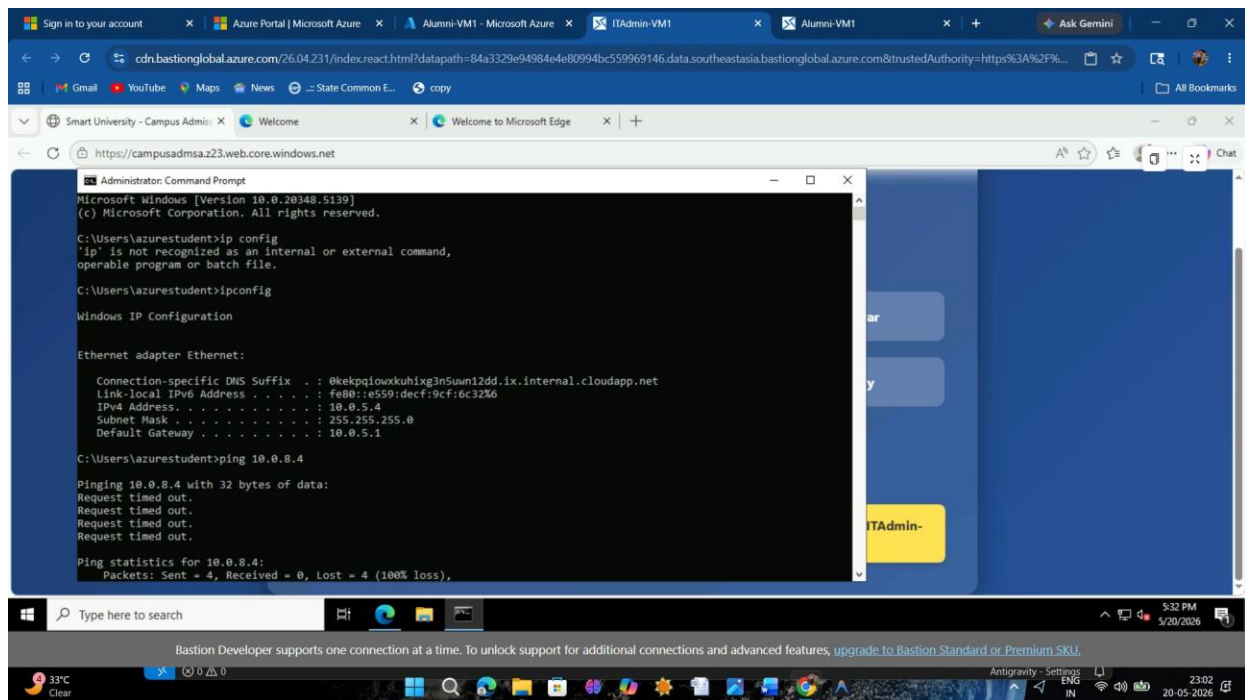
Screenshot 20 — ITAdmin-VM1 — Ping to Alumni Blocked (100% Loss)



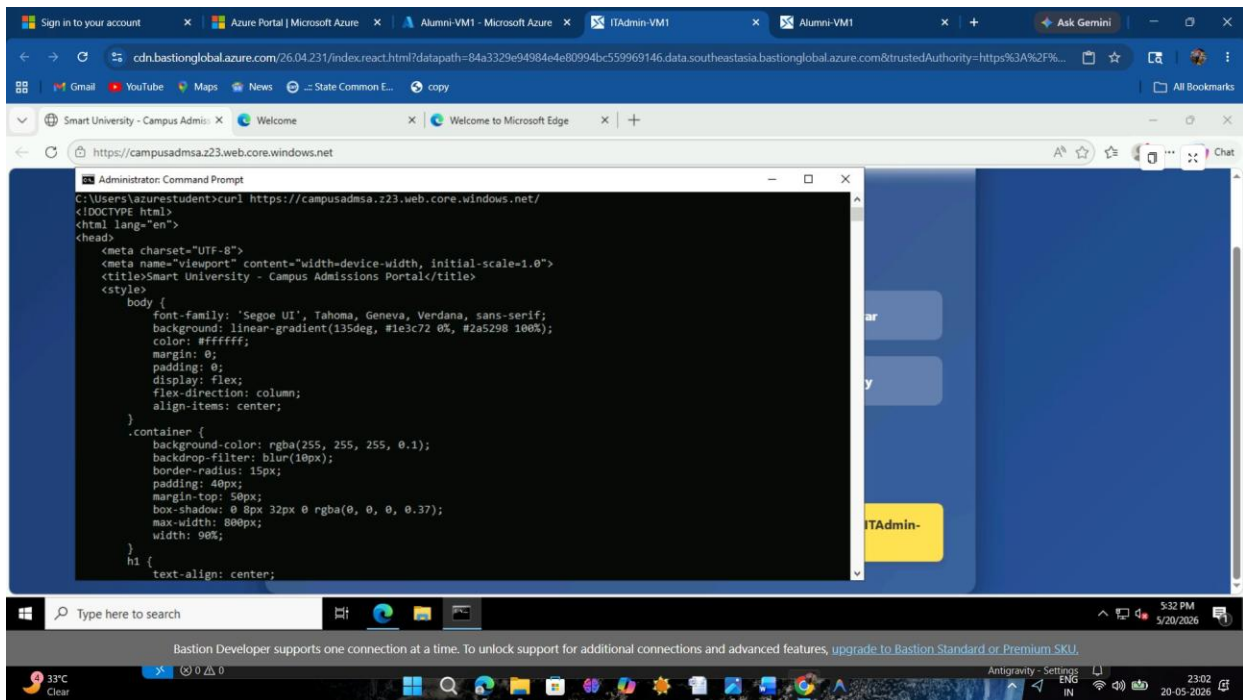
Screenshot 21 — ITAdmin-VM1 — Webpage curl Test



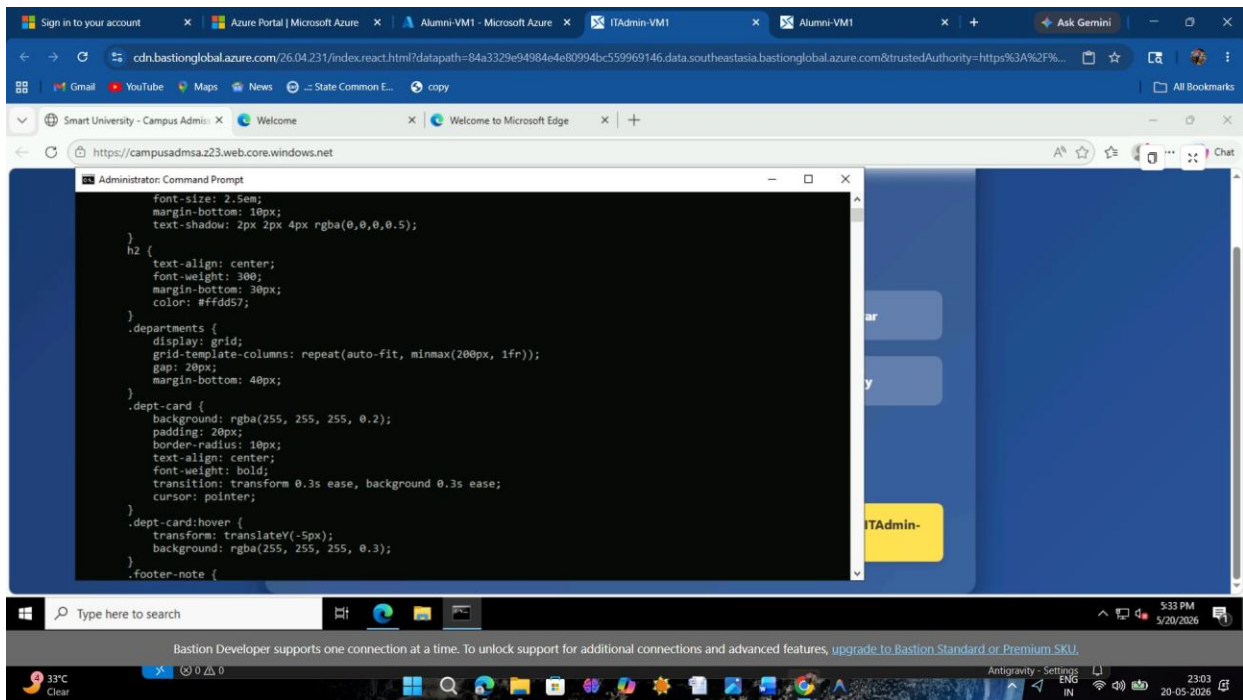
Screenshot 22 — ITAdmin-VM1 — Webpage Launched in Browser



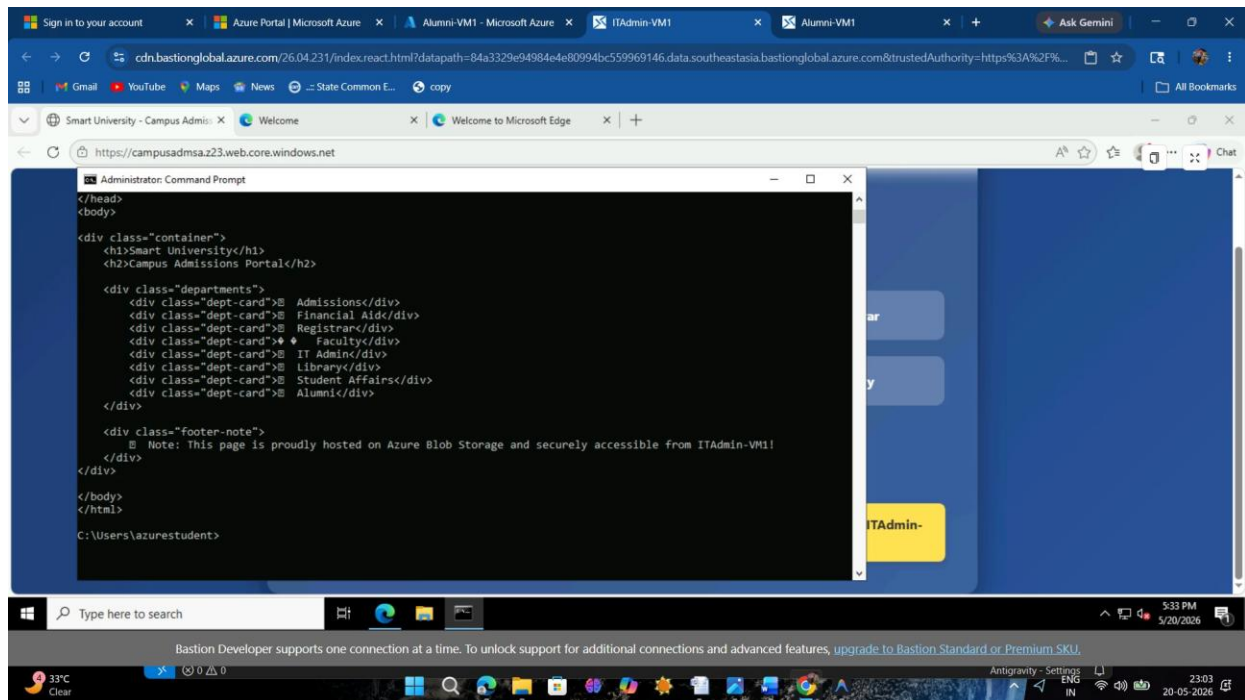
Screenshot 23 — Campus Webpage — Full View in Alumni-VM1 Browser



Screenshot 24 — Webpage with CMD — Both Demo Proofs



Screenshot 25 — Campus Webpage — Clean Full Page View



Screenshot 26 — Alumni-VM1 — Ping to ITAdmin Blocked (Clean View)

